MAT700 Mathematical Methods for Data Mining - Lecture1

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What is Science?

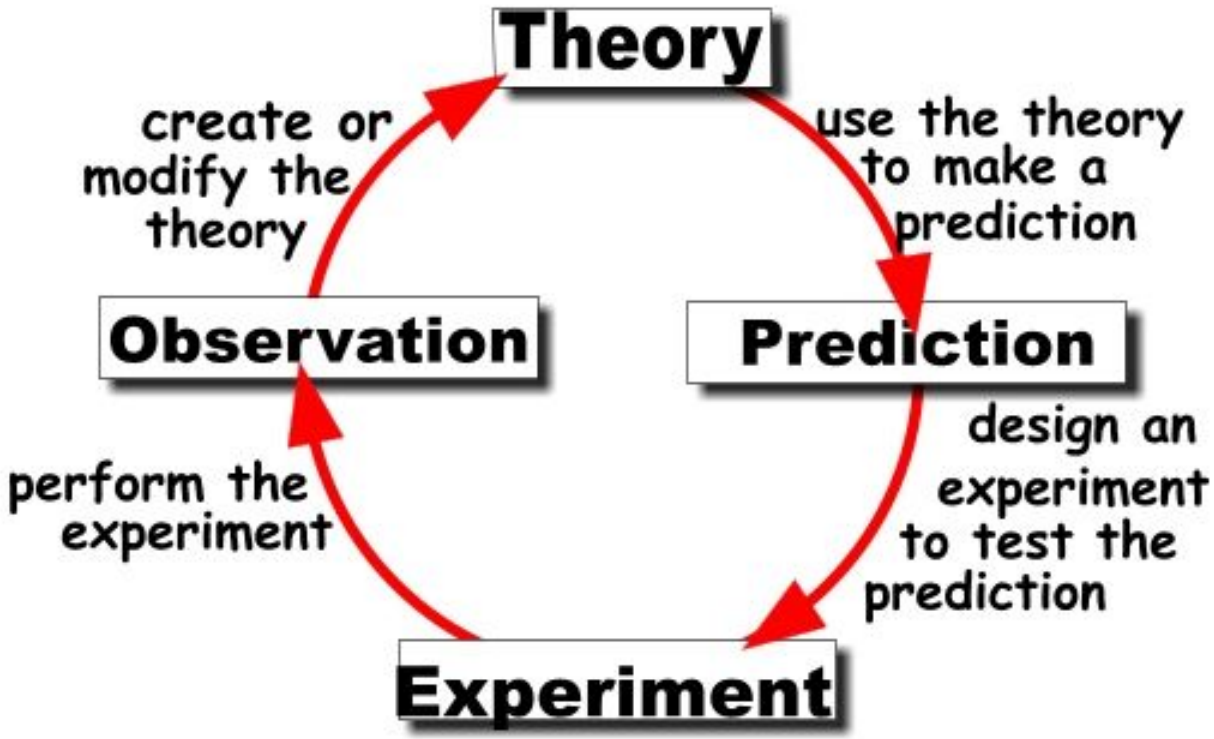
Explain and calculate unknown!



What is Science?

The Scientific Method

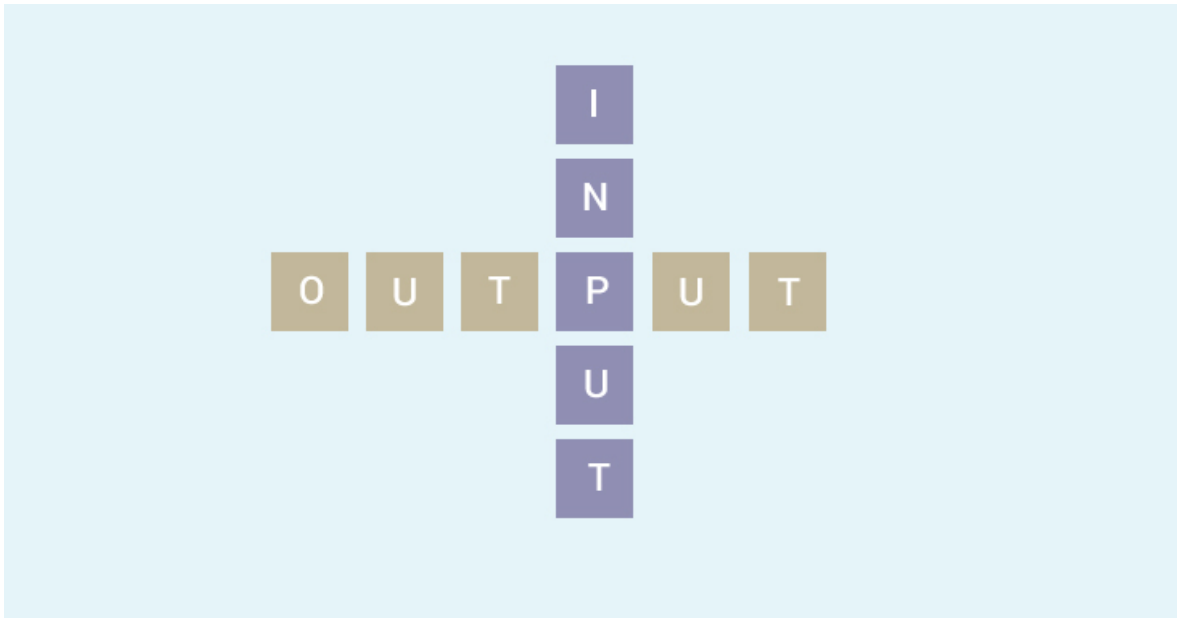
The essence of all good science lies in a process called "the scientific method." The scientific method uses objective experimentation to predict, verify, or refute, an assertion made by a theory. Based on the results of the experiment(s), the theory is modified. The process of predict, verify or refute is repeated, continually testing and modifying the theory until the theory fits all possible experimental observations. Then the theory is considered a scientific law.



What to do if we don't have a good theory?

Answer: Write programmes and solutions from observation!

Any computer programme is a function.



How to calculate an unknown function?

In God we trust, all others bring data.
– William Edwards Deming (1900-1993)

AIMS OF THE MODULE:
To provide an introduction to some mathematical models and methods associated with basic problems in **Data Mining**

Data Mining refers to a set of tools for modeling and understanding complex datasets.

The mathematical models are set up with various mathematical theories, ranging from Bayesian inference approach, optimisation theory, collaborative filtering, cluster and pattern analysis, to kernel methods and learning theory.

1. In this course, we will consider the following problems: classification, cluster and outlier analysis, mining time-series and sequence data, text mining and web mining, ranking, pattern analysis.

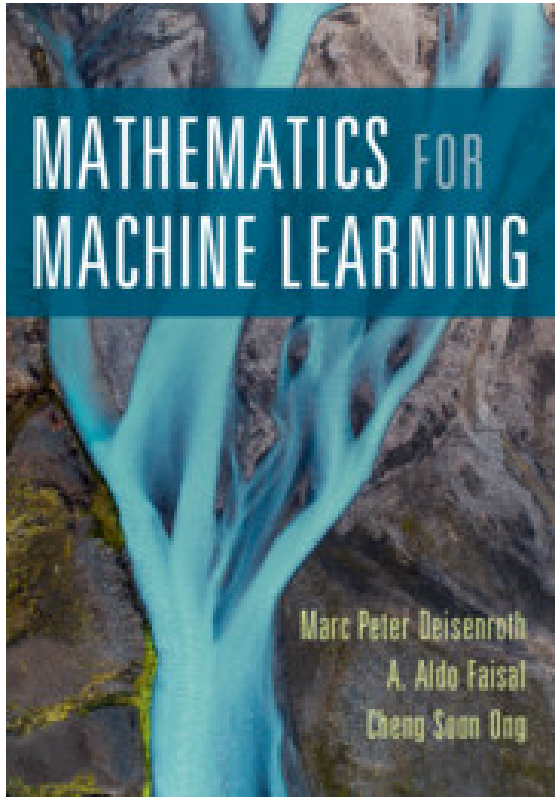
2. METHOD(S) OF SUMMATIVE ASSESSMENT:

   Written Examination: Percentage Contribution to the Module Assessment: 100%

Mathematical Preliminaries

- Linear Algebra
- Calculus
- Probability Theory
- Graph Theory

Recommended Book (free):



The best source of knowledge in Data Science:

kaggle

What is Data Mining?

Data Mining

Data mining is the process of extracting *patterns* from *large data sets* by combining methods from *statistics* and **artificial intelligence** with **database management**.

Results of data mining are *models* or *patterns*.

There are some who regard data mining as synonymous with **machine learning**.

Machine Learning is the ability to teach a computer without explicitly programming it.

Examples are used to train computers to perform tasks that would be difficult to program

Computer Vision: Car detection



Cars

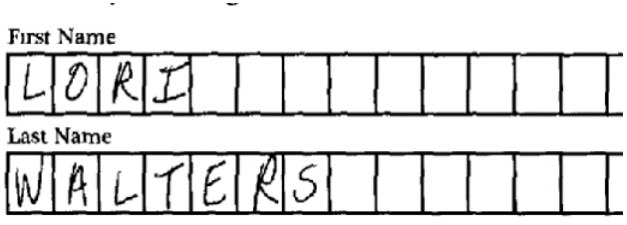


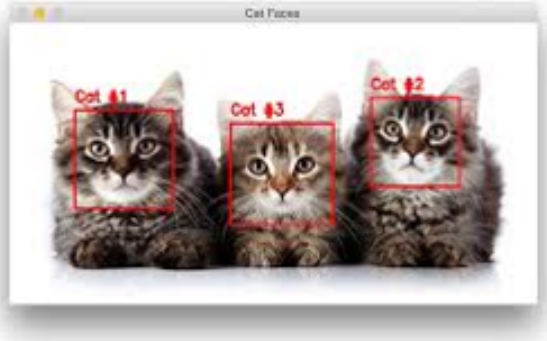
Not a car

Testing:



What is this?





Machine Learning = Learn models from data.
The typical case where machine learning is a good approach is when we have little idea of what we are looking for in the data.

For example, it is rather unclear what it is about movies that make certain movie-goers like or dislike a movie.

On the other hand, machine learning has not proved successful in situations where we can describe the goals of the mining more directly.

An interesting case in point is the attempt by WhizBang Labs to use machine learning to locate people's resumes on theWeb. It was not able to do better than algorithms designed by hand to look for some of the obvious words and phrases that appear in the typical resume.

Famous Applications

- Google
- Amazon
- Netflix
- Email Spam
- Credit Card Fraud Detection
- Automatic Abstracting
- ...

Five Main Types of Applications

Classification, Numerical Prediction, Association, Clustering, Ranking

Statistics vs Data Mining

- Data mining uses data that has usually been collected for some other purpose. Ex. retail data (record of purchases in a store).
- The data mining process may not be connected to the data acquisition process.
- In statistics data is often collected to answer specific questions.

Large Data Sets

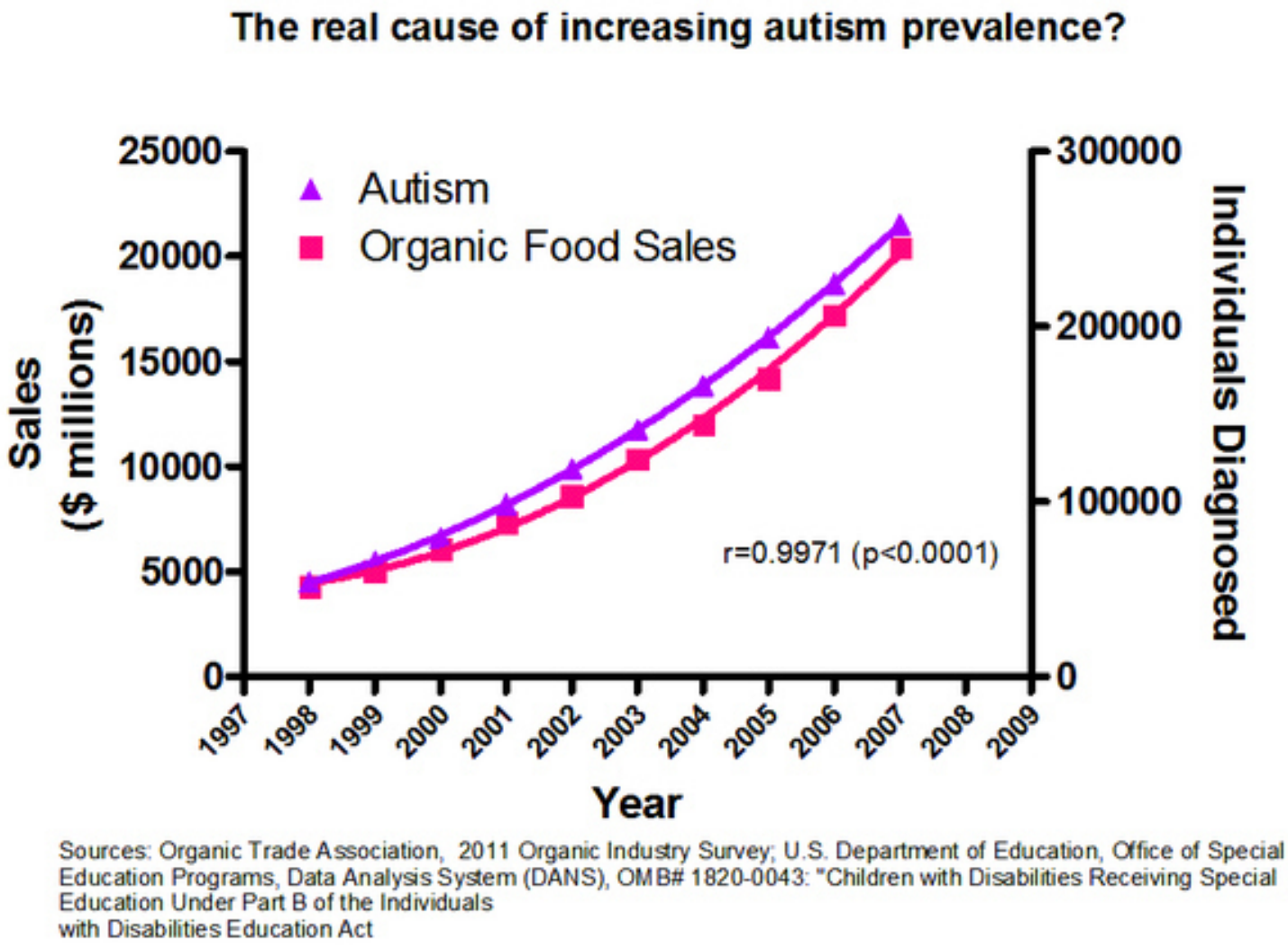
- If data sets were small we would use statistical exploratory analysis.
- Large to a statistician - few hundred to a thousand data points.
  Large to a data miner - millions or billions of data points.
- Exceedingly large datasets have problems associated with them beyond what is traditionally considered by statisticians (e.g., disk access efficiency constraints).

- Many statistical methods require some type of exhaustive search. As datasets become larger wrt to records and variables, search becomes computationally more prohibitive (looking at all subsets of variables requires search of $2^p - 1$ sets)

- Increase in number of variables leads to the *curse of dimensionality*.
  Curse of dimensionality - exponential rate of growth in the number of unit cells in the data space as the number of variables increases.

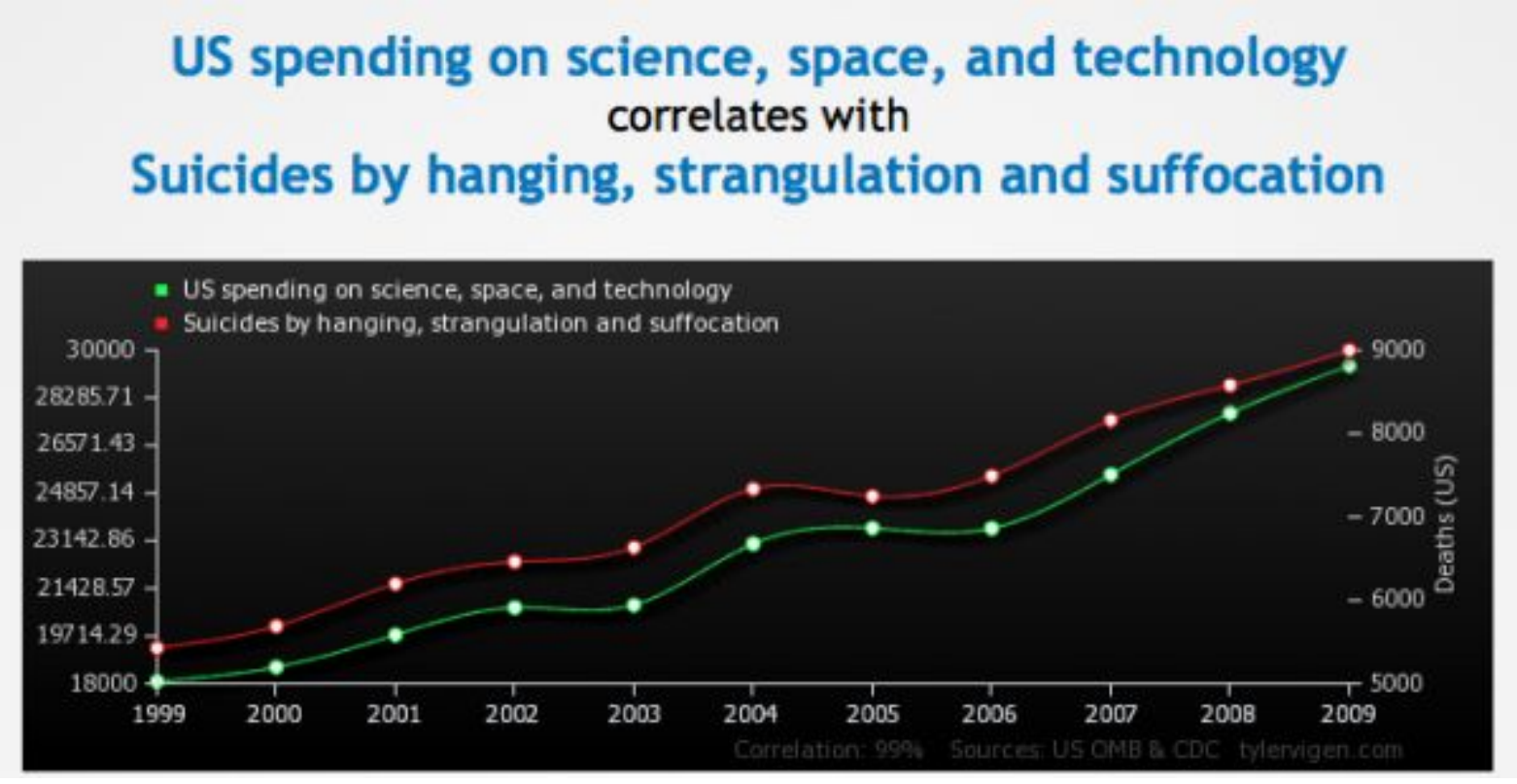
- Given a very large set of variable, $X_1, X_2, X_3, \ldots, X_p$ where p is large, then **there is a high probability** of finding an X that correlates with Y even if no real correlation exists in the domain.

Fake Correlations



Sources: Organic Trade Association, 2011 Organic Industry Survey; U.S. Department of Education, Office of Special Education Programs, Data Analysis System (DANS), OMB# 1820-0043: "Children with Disabilities Receiving Special Education Under Part B of the Individuals with Disabilities Education Act

Fake Correlations



Statistical Limits on Data Mining

Data Mining, noun 1. Torturing the data until it confesses. . . and if you torture it long enough, you can get it to confess to anything.

A common sort of data-mining problem involves *discovering unusual events hidden within massive amounts of data*. This section is a discussion of the problem, including "**Bonferroni's Principle**," a warning against overzealous use of data mining.

Total Information Awareness: In 2002, the Bush administration put forward a plan to mine all the data it could find, including credit-card receipts, hotel records, travel data, and many other kinds of information in order to track terrorist activity.

The concern raised by many is that if you look at so much data, and you try to find within it activities that look like terrorist behaviour, are you not going to find many innocent activities - or even illicit activities that are not terrorism - that will result in visits from the police and maybe worse than just a visit?

The answer is that it all depends on how narrowly you define the activities that you look for. Statisticians have seen this problem in many guises and have a theory, which we introduce in the next slide.

Bonferroni's Principle

Suppose you have a certain amount of data, and you look for *events of a certain type* within that data.

You can expect events of this type to *occur*, even if the data is *completely random*, and the number of occurrences of these events will grow as the size of the data grows.

These occurrences are "*bogus*," in the sense that they have no cause other than that random data will always have some number of unusual features that look significant but aren't.

Without going into the statistical details, we offer

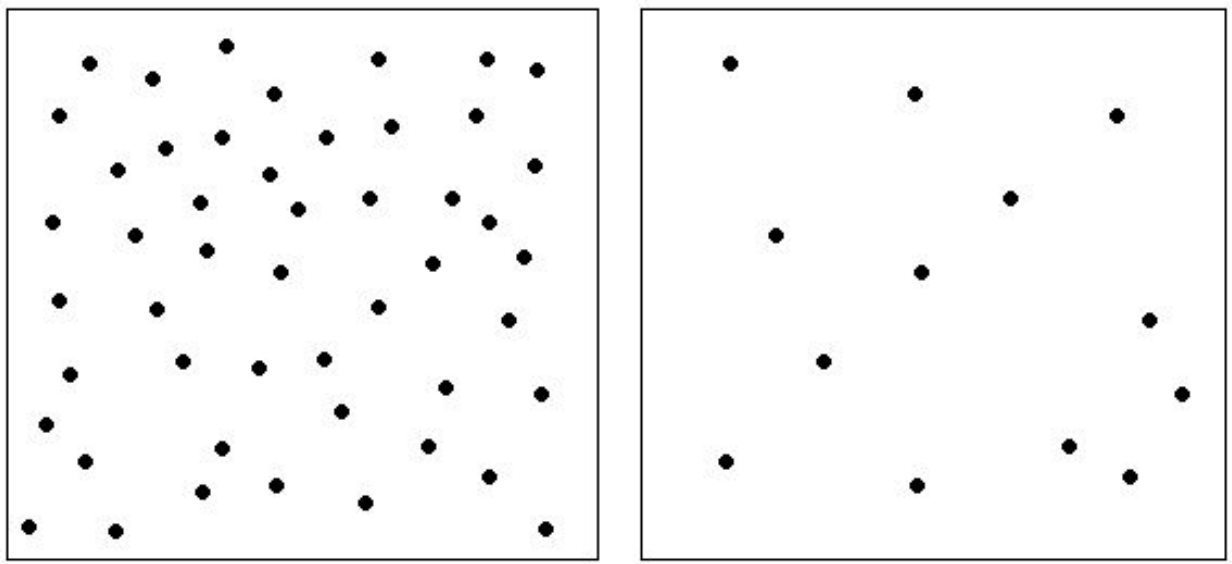
An informal version of Bonferroni's principle

Calculate the expected number of occurrences of the events you are looking for, on the assumption that data is random.

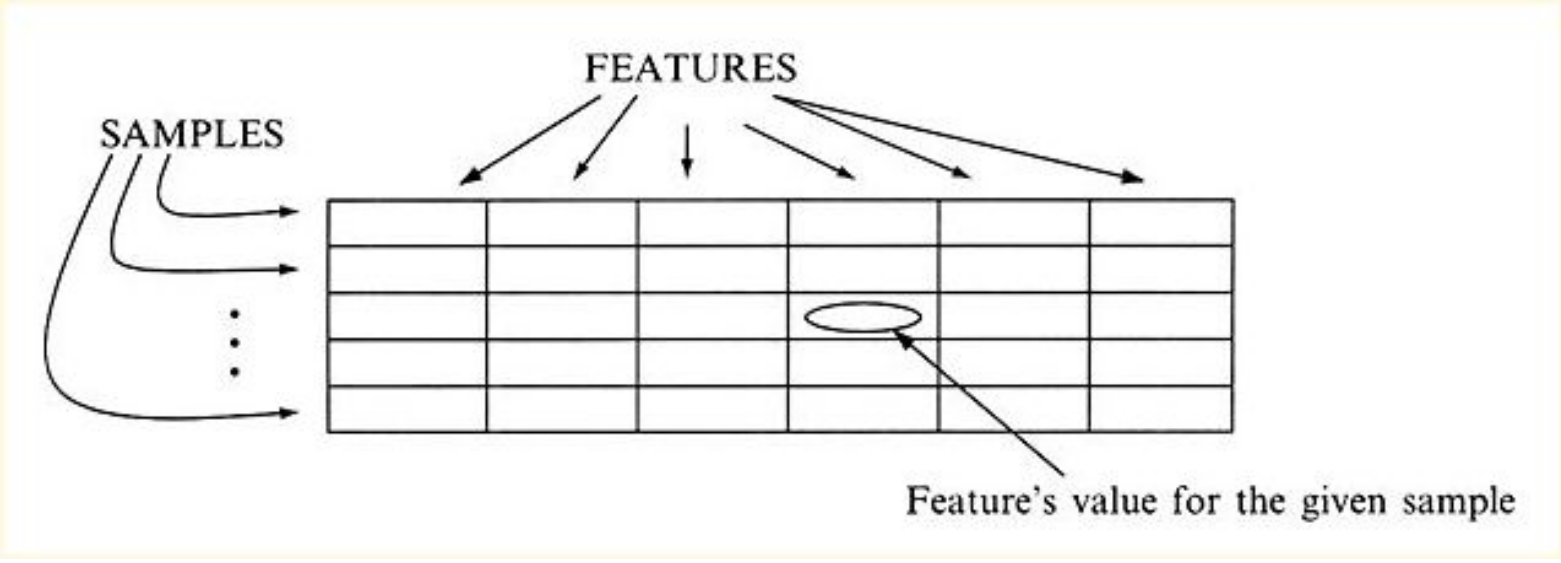
If this number is *significantly larger* than the number of real instances you hope to find, then you must expect almost anything you find to be bogus, i.e., a *statistical artifact* rather than evidence of what you are looking for.

Examples

- Birthday Paradox: $E(M_2) = \frac{K(K-1)}{2} \times \frac{1}{N}$.
- The Helmholtz principle in human perception:



Tabular Representation of a Data Set



In general we have a dataset of examples (called *instances or samples*), each of which comprises the values of a number of variables, which in data mining are often called *attributes or features*.

Labelled and Unlabelled Data

There are two types of data, which are treated in radically different ways.

- For the first type there is a **specially designated attribute** and the aim is to use the data given to predict the value of that attribute for instances that have not yet been seen.
  Data of this kind is called *labelled*.
  Data mining using labelled data is known as *supervised learning*.

- If the designated attribute is *categorical*, i.e. it must take one of a number of distinct values such as "very good", "good" or "poor", or (in an object recognition application) "car", "bicycle", "person", "bus" or "taxi" the task is called *classification*.

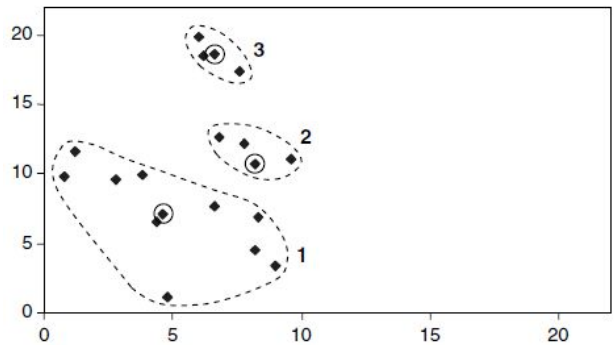
- If the designated attribute is numerical, e.g. the expected sale price of a house or the opening price of a share on tomorrow's stock market, the task is called *regression*.

Data that does not have any specially designated attribute is called *unlabelled*.
Data mining of unlabelled data is known as *unsupervised learning*.
Here the aim is simply to extract the most information we can from the data available.

Example (Association Rules)

IF cheese AND milk THEN bread (probability = 0.7)

Example (Clustering)



Outlier Analysis

A data set may contain objects that do not comply with the general behaviour or model of the data (e.g., "Napoleon Dynamite" in the Netflix Competition).
These data objects are **outliers**. Many data mining methods discard outliers as noise or exceptions.

However, in many applications (e.g., fraud detection, text mining, social networks analysis, ...) the rare events can be more interesting than the more regularly occurring ones.

The analysis of outlier data is referred to as **outlier analysis** or **anomaly mining**.

Outliers may be detected using statistical tests that assume a distribution or probability model for the data, or using distance measures where objects that are **remote** from any other cluster are considered outliers.

Example (Outlier analysis)

Outlier analysis may uncover fraudulent usage of credit cards by detecting purchases of unusually large amounts for a given account number in comparison to regular charges incurred by the same account. Outlier values may also be detected with respect to the locations and type of purchase, or the purchase frequency.

Data Preparation

Although this course is about data mining not data preparation, some general comments about the latter may be helpful.

- **Data Cleaning**: Even when the data is in the standard form it cannot be assumed that it is error free. In real-world datasets erroneous values can be recorded for a variety of reasons, including measurement errors, subjective judgements and malfunctioning or misuse of automatic recording equipment.

- **Missing Values**: In many real-world datasets data values are not recorded for all attributes.

Reducing the Number of Attributes

Although it is tempting to store more and more information about each instance (especially as it avoids making hard decisions about what information is really needed) it risks being self-defeating.

Suppose we have 10,000 pieces of information about each supermarket customer and want to predict which customers will buy a new brand of dog food. The number of attributes of any relevance to this is probably very small.

At best the many irrelevant attributes will place an unnecessary computational overhead on any data mining algorithm. At worst, they may cause the algorithm to give poor results.

There are several ways in which the number of attributes (or "features") can be reduced before a dataset is processed. The term *feature reduction* or *dimension reduction* is generally used for this process.